Two-Hop Fact-Based Reasoning (done in main doc)

Math Fact-Based Reasoning

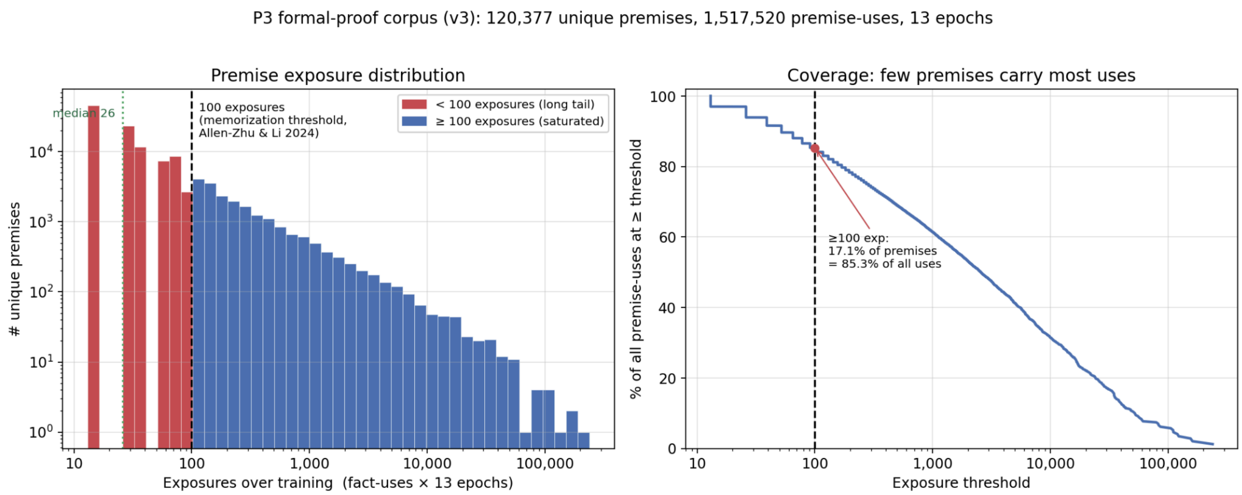
Research Question and Hypothesis

While two-hop reasoning is fact-based, our implementation remains reducible to a fact-agnostic solving recipe, as it is the deterministic composition of two sequential queries. As such, we intend to evaluate a stronger version of the same question regarding whether a split model can perform fact-based reasoning: by being responsible for mathematical reasoning from a given block of premises. Unlike two-hop (which has a fact-invariant general approach), we theorize that the path taken to do a mathematical proof notably changes on the basis of the axiomatic "facts" at hand. By making the problem space contingent on facts, the model will be more directly tested on fact-based reasoning in the stricter sense, rather than as more generally understood. Furthermore, unlike previous experiments which attempt to implement a retriever system to aid the model, we instead assume a "perfect" retriever which, unlike LMLM or Co-LMLM, doesn't even need to be queried to know exactly what facts are required to solve the problem. By doing so, the problem of retriever optimality can be abstracted away and the pure viability of offloading facts to an entirely separate system from the LLM can be evaluated (Zhao et al., 2025, Feldman et al., 2026).

Experimental Setup and Dataset

To successfully maintain a discriminant between the split and dense model, instead of pre-training a model from scratch and risking both the split and dense model achieving 0% on the mathematical problems from poor initial generalization, we instead take the final trained checkpoint of Qwen2.5-0.5B base, which successfully generated ~50% (NOTE: ask Tejas for the exact stat) of next tokens and 0% of entire valid proofs under teacher forcing and then do supervised fine-tuning of both the split and dense models. Qwen 2.5-0.5B is already trained on some mathematical reasoning, which means that both the split and dense model technically have some mathematical facts already, but given the largely synthetic naming scheme of proofs and relatively low preexisting success rate this was an accepted point of potential further clarification.

The dataset was tokenized using the Qwen2.5 tokenizer for maximum compatibility. Mathematically formal proofs were compiled from Metamath's set.mm, iset.mm, and nf.mm, Mizar, Magnushammer (a solved set from Isabelle), thproofs, ATP/E-prover prf2, and distinct Mizar alternate runs from ENIGMA mzr 01/02/03/08 for a total of 467.2 million tokens, repeated for 13 epochs in order to achieve a total of 6.07 billion tokens seen by the model. 17.08% of unique premises receive 100 exposures, which is by Knowledge Parameter Scaling Laws sufficient for somewhat memorizing the premises, for a total of 85.28% of premise uses in the data (Allen-Zhu and Li, 2024). Our math proofs are organized as premise blocks starting with “I know these mathematical statements:” followed by the list of factual premises (as well as a list of local assumptions), and then a “----” to indicate the start of the actual proof, which has starts with a supervised “GOAL” line that the split and dense models target.



Conceptually, the setup during training is similar to Co-LMLM by wrapping a fact by a special query sequence, but unlike Co-LMLM the split model the wrapper around the factual premise block is also masked out in Co-LMLM style, as it is assumed that the split model doesn’t even need to know what or when to query for the perfect retriever to gather and return all necessary info to solve the problem (Feldman et al., 2026). Dense, by contrast, also receives the trace with facts provided but does not mask them out, instead including the facts in the loss function and encouraging memorization. During gradient backpropagation, rather than dividing the total gradient by the number of supervised tokens (which might fairly be interpreted as equivalent to increasing learning rate for the reasoning section), we instead divide by a fixed 262, 144 constant to represent the global batch size (which is the number of supervised tokens per batch for dense and number of total tokens per batch for split). Through this, essentially only the reasoning section of the gradient is preserved at similar ratios to the reasoning section of the dense, rather than being multiplied. While multiplying would likely benefit split, the goal was to isolate the “noise,” so to speak, created by tuning for data.

Evaluations

During training, cross-entropy loss and supervised token fraction were monitored. The split model has loss on fewer tokens than the dense model which is also trained to memorize the fact block, so loss was expected to be lower for the split model without any particular meaning.

During the evaluation runs, a held out set of proofs, facts_present, from each family were used with facts in the premise block that the models have never seen before anywhere in a mathematical trace either as premises, intermediates, or the goal. Both models are run over the entire set of four thousand held out traces with the premise block given. Additionally, for a deterministic 10% of reasoning traces (facts_absent) the two models must also try to solve it without being given a premise block to verify the split does not learn any facts and the dense model learns facts, and for 10% (facts_corruption) statements of premises are changed to measure split and dense generalization. For all three categories facts_present, facts_absent, and facts_corruption, cross-entropy loss and token prediction accuracy is measured using teacher forcing. The Metamath proofs for facts_present are also all deterministically verified using the Metamath tri-state solver, which categorizes them into valid, invalid, or unknown. For the run with the premise block, teacher forcing is used to give a percent of tokens generated correctly, and for the metamath sub-section of the eval the proof generated by the split and dense models is also deterministically verified. There are 4191 eval rows used for a total of 7.041 million tokens (8.46 million including the 10% diagnostics) and Qwen2.5-0.5B base is run on the same eval set as a base benchmark.

Results

When the correct premise block is given to both models, the split model performs better than dense on every end-to-end Metamath endpoint, while Qwen2.50.5B base does not generate a successful proof under any of them. Furthermore, the confidence intervals for each splitdense difference remain entirely above zero, suggesting that the advantage is not dependent on a small number of unusually easy proofs for split. While the absolute amount of wholeproof success remains relatively low, the agreement between exact match and deterministic validity means the difference cannot be explained only by the split model producing a different but still invalid proof form. As such, when access to factual premises is held equal, removing the responsibility of memorizing those premises does not harm mathematical reasoning and instead appears to somewhat improve it

Endpoint	**base**	**dense**	**split**	**Δ split−dense [95% CI]**
Exact match (evaluated)	0.00%	7.89%	10.93%	+3.04% [+1.21, +5.06]
Exact match (budget-eligible)	0.00%	8.44%	11.69%	+3.25% [+1.30, +5.41]
Metamath validity (decided)	0.00%	9.11%	12.35%	+3.24% [+1.21, +5.26]

`facts_absent` (header only) — split − dense:

Family	n	Base NLL (acc)	Dense NLL (acc)	Split NLL (acc)	ΔNLL split−dense [95% CI]	ΔAcc split−dense [95% CI]	Gold paste
enigma	27	0.3080 (91.29%)	0.0780 (97.84%)	0.0835 (97.88%)	+0.0055 [+0.0036, +0.0077]	+0.04% [−0.03, +0.10] (CI incl. 0)	76.2%
isabelle	59	0.7679 (83.78%)	0.4565 (88.92%)	0.4462 (89.39%)	−0.0103 [−0.0206, −0.0002]	+0.46% [+0.20, +0.80]	72.4%
metamath	50	0.5020 (88.87%)	0.2910 (93.22%)	0.3720 (92.38%)	+0.0811 [+0.0676, +0.0987]	−0.83% [−1.03, −0.69]	78.6%
mizar	249	1.0685 (73.57%)	0.7197 (80.06%)	0.7527 (79.86%)	+0.0330 [+0.0256, +0.0421]	−0.20% [−0.33, −0.07]	29.3%
prf2	32	0.3312 (90.59%)	0.0883 (97.60%)	0.0972 (97.56%)	+0.0089 [+0.0064, +0.0118]	−0.04% [−0.10, +0.01] (CI incl. 0)	77.8%
thproofs	5	0.9615 (78.39%)	0.6666 (83.80%)	0.6850 (84.20%)	+0.0184 [+0.0061, +0.1249]	+0.40% [−0.27, +0.68] (CI incl. 0)	21.7%

When the premise block is removed, the direction of the result generally reverses, with split receiving a worse likelihood than dense for nearly every family and the strongest loss in next token accuracy occurring on Metamath. This is the behavior that should occur if dense has stored part of the premise content in its weights while split has learned to depend more heavily on the externally supplied facts. The Isabelle exception, as well as accuracy differences that are close to zero for several families, means this should not be interpreted to show that split learns no mathematical facts at all. Instead, when considered alongside the facts_present condition, the result demonstrates that a greater portion of split's performance is contingent on access to the premise block, and the base Qwen model likely has some pre-existing knowledge of the facts in the dataset.

`facts_corrupted` (names kept, statements swapped) — split − dense:

Family	n	Base NLL (acc)	Dense NLL (acc)	Split NLL (acc)	ΔNLL split−dense [95% CI]	ΔAcc split−dense [95% CI]
enigma	27	0.3446 (90.29%)	0.0793 (97.69%)	0.0883 (97.71%)	+0.0090 [+0.0062, +0.0122]	+0.02% [−0.06, +0.09] (CI incl. 0)
isabelle	59	1.0054 (79.37%)	0.6061 (85.75%)	0.6105 (85.69%)	+0.0044 [−0.0061, +0.0189] (CI incl. 0)	−0.05% [−0.28, +0.18] (CI incl. 0)
metamath	50	0.3741 (90.43%)	0.2533 (93.63%)	0.3074 (92.88%)	+0.0541 [+0.0459, +0.0648]	−0.76% [−0.93, −0.62]
mizar	249	1.0065 (75.12%)	0.6111 (82.66%)	0.6146 (82.80%)	+0.0035 [+0.0007, +0.0066]	+0.14% [+0.05, +0.24]
prf2	32	0.3186 (90.69%)	0.0795 (97.62%)	0.0851 (97.70%)	+0.0057 [+0.0035, +0.0082]	+0.08% [+0.03, +0.12]
thproofs	5	1.3109 (69.77%)	0.7563 (80.18%)	0.7896 (80.23%)	+0.0332 [+0.0102, +0.0686]	+0.05% [−0.59, +0.65] (CI incl. 0)